

Individual Essay Assignment

Overview

The purpose of this project is to consider a contemporary commercial or public infrastructure application of technology through the lenses of Society and Sustainability. To be considered contemporary, commercial projects or products must be either in construction / development or active commercial operation (ie making money). Public infrastructure projects must be in construction or active service, past projects now closed and infrastructure maintained as a historic site (such as the original infrastructure of the Trent-Severn Waterway) may not be used.

For your project you are to:

- provide a background on the technology and the application
- identify the social concerns (including disproportionate impacts on equity deserving groups) that arise from the specific application, as well as any mitigations in place and a critique of those
- identify the sustainability / environmental concerns that arise from the specific application, as well as any mitigations in place and a critique of those

Depending on the topic selected it is possible that one area (social or sustainability / environmental) will be predominant. If this is the case for your project, be aware that all areas contribute to your grade and strengthen your analysis as needed.

Take a 'life cycle' approach to your application, considering the project from resource extraction through to end of life disposal / recycling.

Deliverables

Topic Selection

To ensure that topics are not duplicated, all students must submit their preferred specific application as well as alternate choices. Topics will be assigned based on student preference, with an aim to reducing the duplication of project themes.

Proposal

To assist students in submitting a quality final product, students will submit a 1 to 2 page proposal for feedback. The proposal should contain the following:

- A brief background on the technology;
- Lists of identified social and sustainability / environmental concerns;
- A preliminary reference list.

At the proposal stage you should have identified approximately 50% of the concerns and 30% of your references.

Progress Report

Similarly to the Proposal, the Progress Report is done to ensure that students work is on track with the project goals. The Progress Report should contain:

- Lists of social and sustainability / environmental concerns with preliminary analysis;
- A refined reference list.

At the progress report stage you should have identified nearly all of the concerns and 75% of your references.

Essay

The Essay / Report should provide an overview of the technology used in your application as well as details of the application and the specific implementation. The Essay should present the identified concerns and mitigations as well as your critique of those.

The Essay is to be in a formal style, including a title page, table of contents and appropriate lists. The body of the essay is limited to a maximum of 10 pages as determined by page numbering (12pt font, single spaced with 1" margins). You may place large figures, tables etc. in an appendix.

Assume that the reader has the technical background of an upper year engineering student. It should be clear to your reader which sections of the essay are facts and which are critique.

The Essay is to be researched thoroughly, drawing from multiple high quality sources.

Students are strongly encouraged to include academic and engineering society databases, as well as government / NGO publications in their research and not rely solely on web search results. Use a referencing / citation style that is appropriate to your discipline.

Peer Teach Presentation

The peer teach presentation is to be approximately 20 minutes in length and should provide the class with a background on the technology, the application as well as the concerns identified. The peer teach should be presented with a neutral bias.

Peer Teach Handout

The handout to accompany the peer teach presentation should provide students with a take home of key information from the presentation. Handouts are due the Wednesday before your scheduled presentation and will be posted for the class.

Deadlines and Evaluation Plan

Deliverable	Due Date	Grade
Topic Selection	September 25	
Proposal	October 16	2.5%
Progress Report	November 6	2.5%
Essay	December 4	30%
Peer Teach Presentation	December 3 or 10 (as scheduled)	15%
Peer Teach Handout	Monday before your scheduled presentation	10%

Rubrics for all deliverables will be posted on SLATE. Other than the presentation, all deliverables are to be submitted via the appropriate assignment on SLATE and are due at 5:00pm. A detailed presentation schedule will be posted in November.